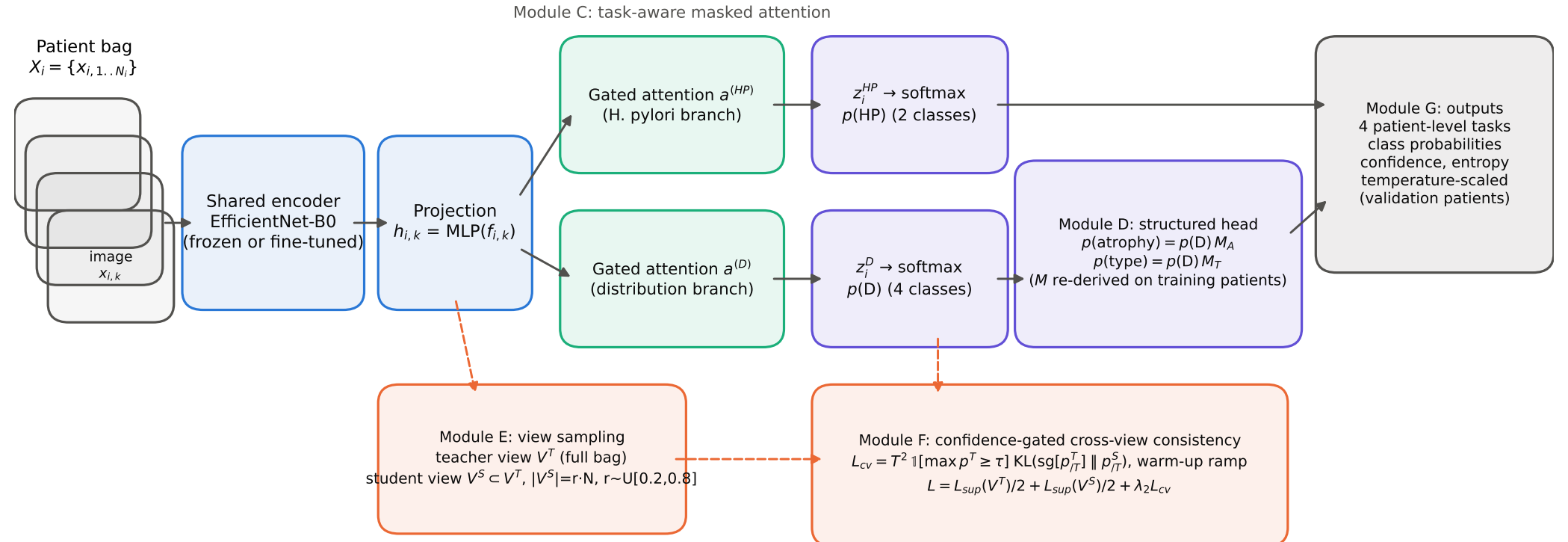
Module C: task-aware masked attention
Patient bag
$X_i = \{x_{i,1..N_i}\}$
image
$x_{i,k}$
Shared encoder
EfficientNet-B0
(frozen or fine-tuned)
Projection
$h_{i,k} = \mathrm{MLP}(f_{i,k})$
Gated attention $a^{(HP)}$
(H. pylori branch)
$z_i^{HP} \rightarrow$ softmax
p(HP) (2 classes)
Gated attention $a^{(D)}$
(distribution branch)
$z_i^{D} \rightarrow$ softmax
p(D) (4 classes)
Module D: structured head
$p(\text{atrophy}) = p(\text{D})\, M_A$
$p(\text{type}) = p(\text{D})\, M_T$
(M re-derived on training patients)
Module G: outputs
4 patient-level tasks
class probabilities
confidence, entropy
temperature-scaled
(validation patients)
Module E: view sampling
teacher view V^T (full bag)
student view $V^S \subset V^T$, $|V^S|$=r·N, r~U[0.2,0.8]
Module F: confidence-gated cross-view consistency
$L_{cv} = T^2\, \mathbb{1}[\max p^T \geq \tau]\, \mathrm{KL}(\mathrm{sg}[p_{/T}^T] \,\|\, p_{/T}^S)$, warm-up ramp
$L = L_{sup}(V^T)/2 + L_{sup}(V^S)/2 + \lambda_2 L_{cv}$